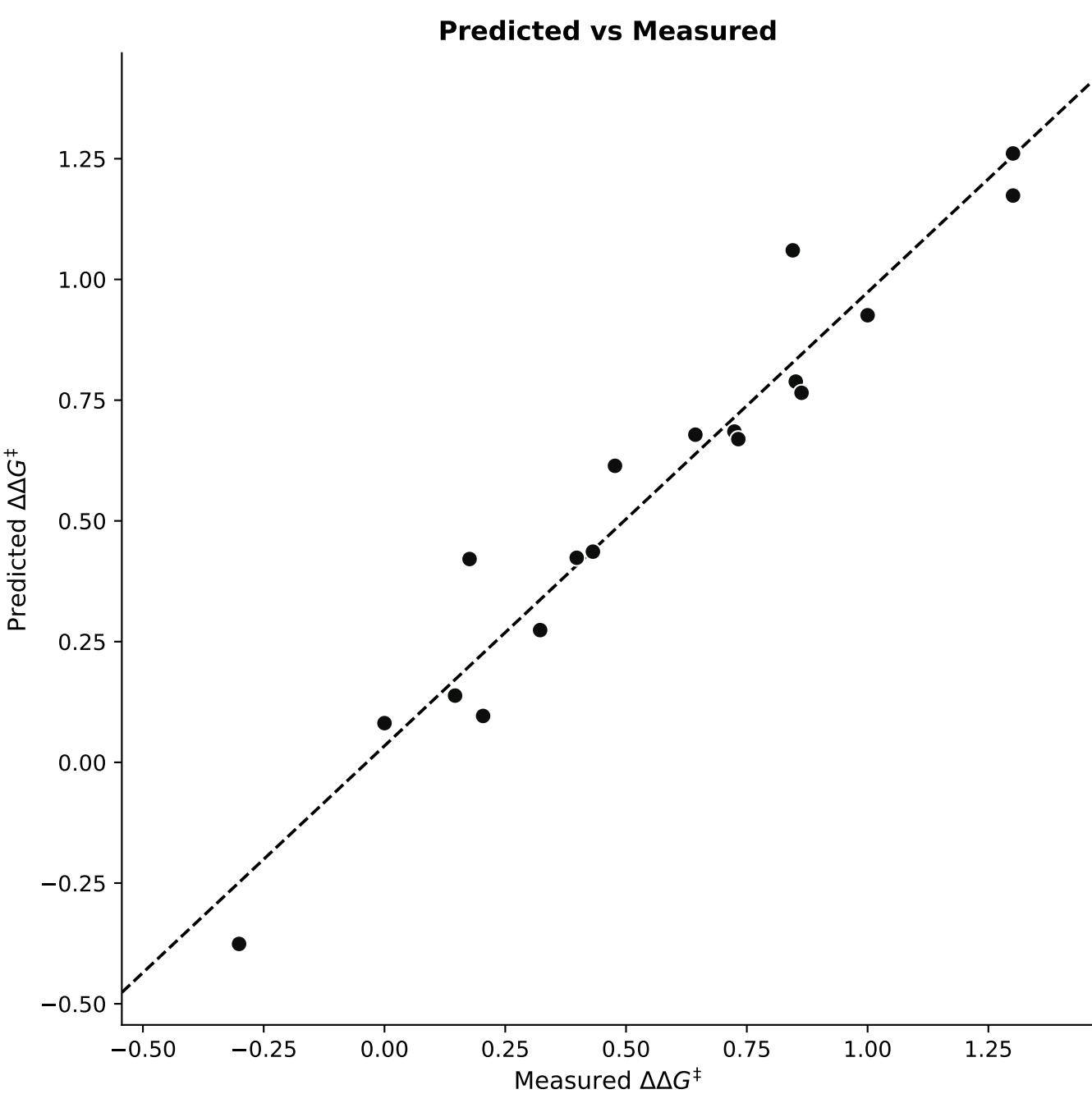




Model Metrics

Metric	Value
R ² (train)	0.9392
adj R ² (train)	0.9262
MAE (train)	0.0827
Q ² 3-fold	0.8790
MAE 3-fold	0.1153
Q ² 5-fold	0.8969
MAE 5-fold	0.1069
Q ² LOOCV	0.8863
MAE LOOCV	0.1110

Coefficient Estimates

Estimate	Std. Error	t value	p value
0.562	0.028	20.169	0.0
0.189	0.034	5.628	0.0
0.418	0.04	10.421	0.0
0.241	0.039	6.123	0.0

Cross-Validation Metrics

Metric	3-Fold	5-Fold	LOOCV
Q²	0.879	0.8969	0.8863
MAE	0.1153	0.1069	0.111
RMSD	0.1462	0.1358	0.1426

Variance Inflation Factors (VIF)

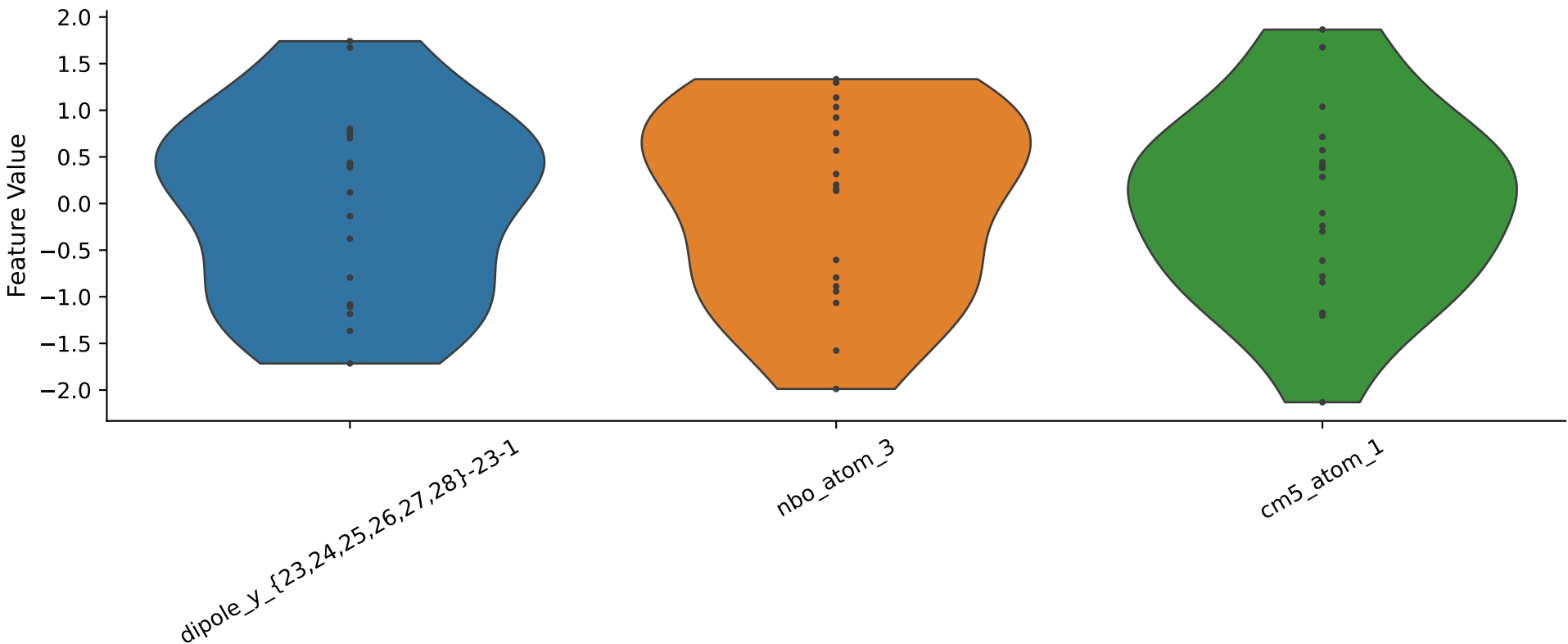
variable	VIF
nbo_atom_3	2.076
cm5_atom_1	1.996
dipole_y_{23,24,25,26,27,28}-23-1	1.455

Coefficient Estimates

Estimate	Std. Error	t value	p value
0.562	0.0279	20.1694	0.0
0.1891	0.0336	5.6278	0.0001
0.4184	0.0401	10.421	0.0
0.2411	0.0394	6.1233	0.0

No external validation set

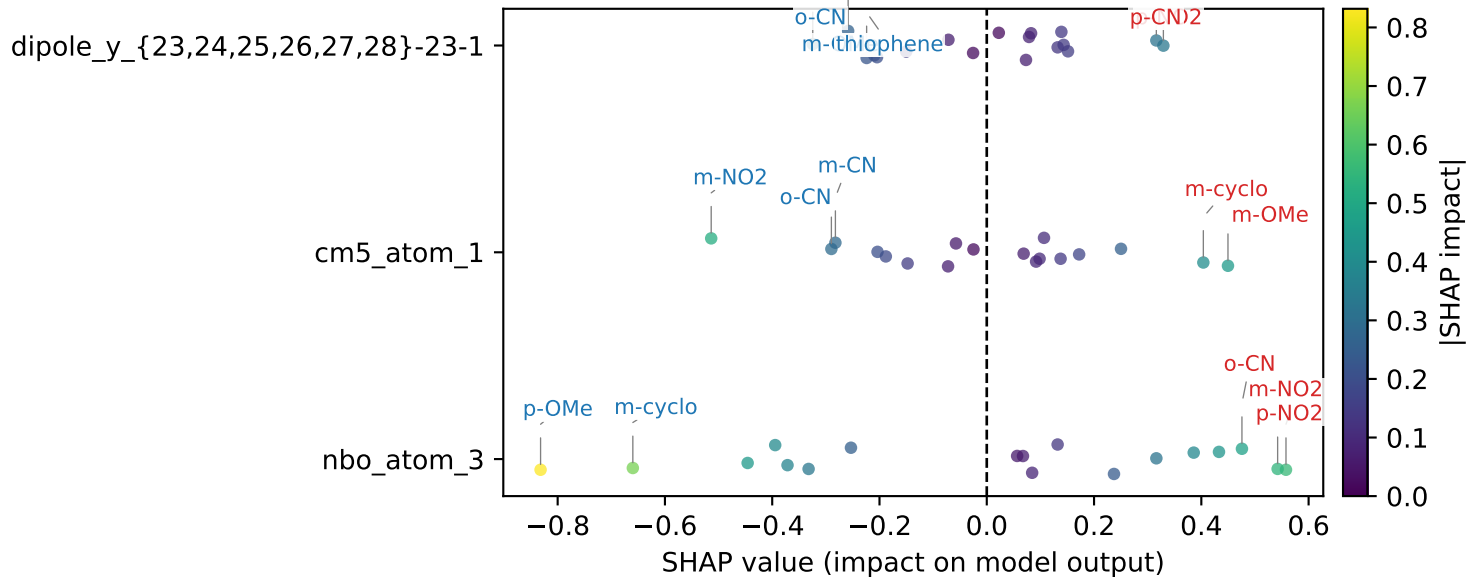
Distribution of Selected Features



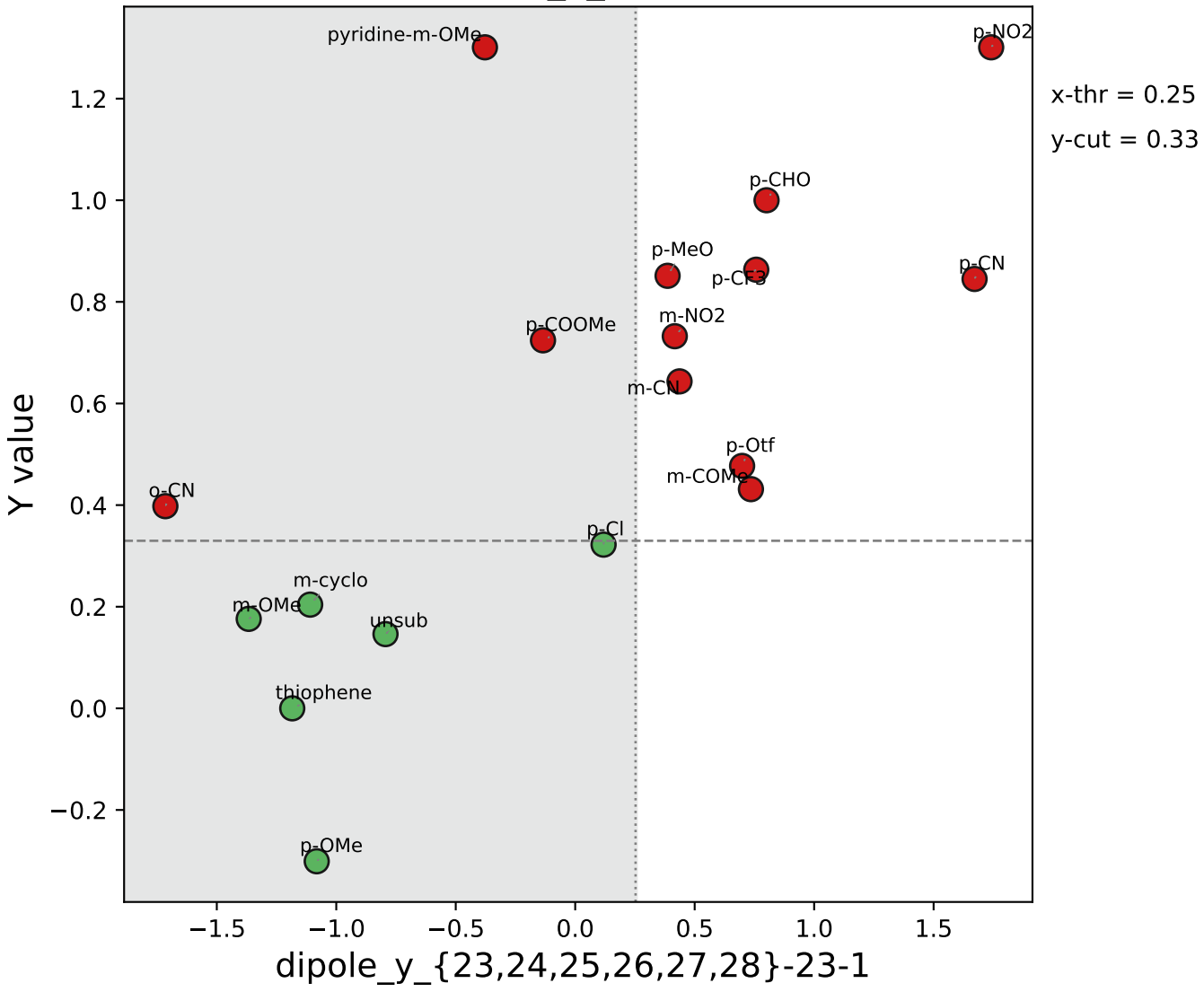
Descriptive Statistics

Feature	mean	std	min	25%	50%	75%	max
dipole_y_{23,24,25,26,27,28}-23-1	-0.0	1.029	-1.715	-1.01	0.253	0.726	1.741
nbo_atom_3	0.0	1.029	-1.988	-0.864	0.182	0.881	1.333
cm5_atom_1	-0.0	1.029	-2.131	-0.739	0.091	0.539	1.865

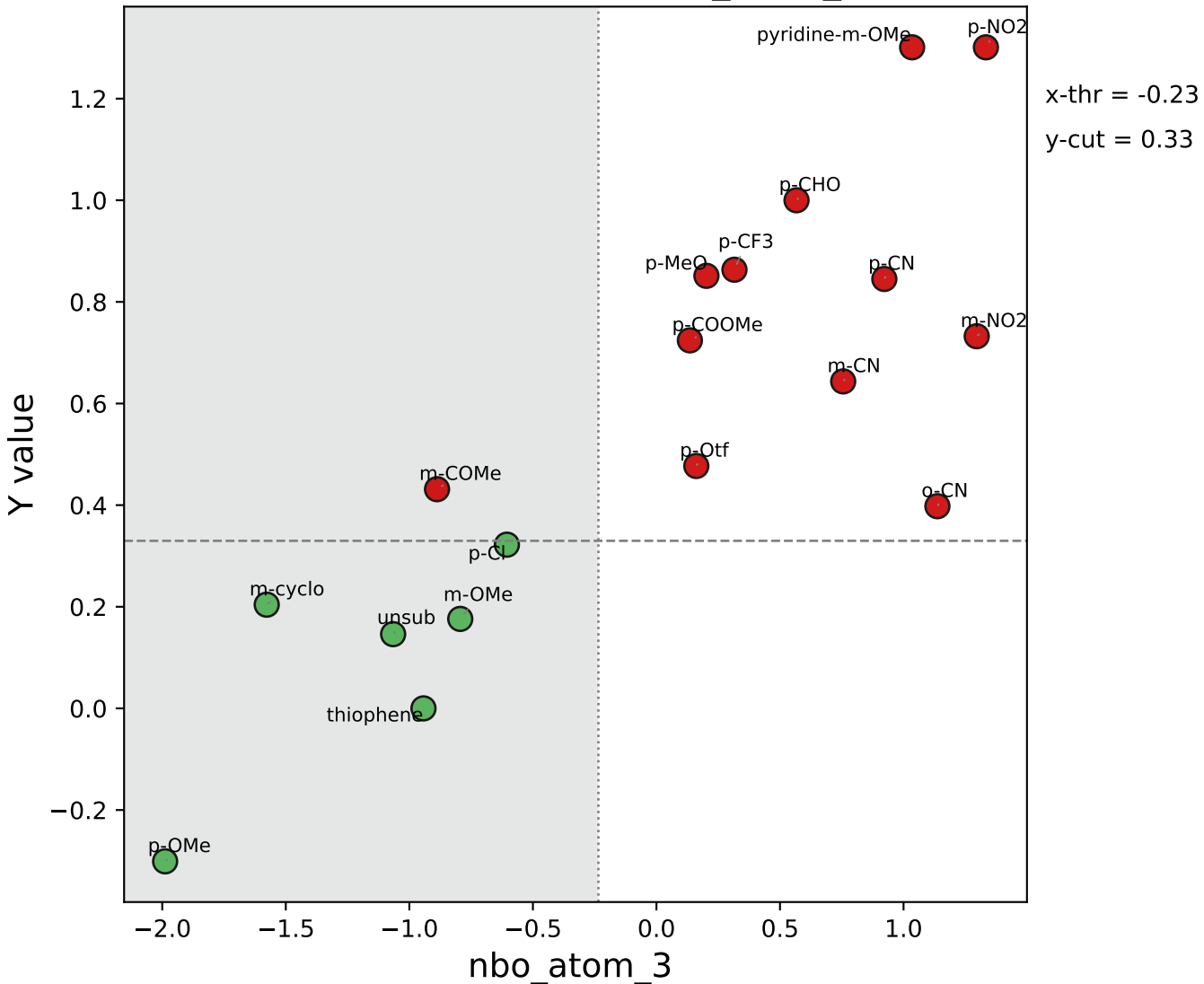
SHAP beeswarm — top 3 features (target)



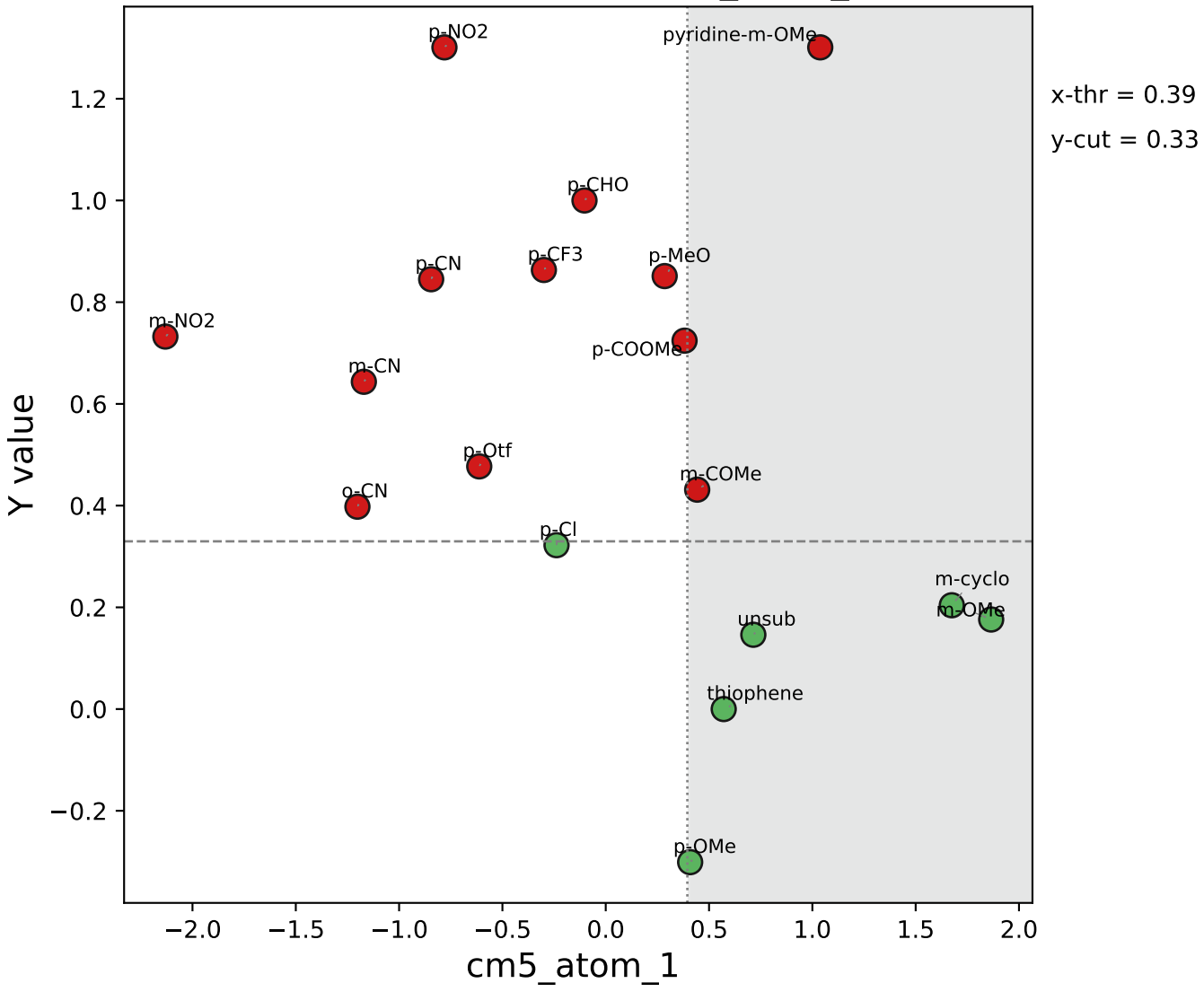
Threshold analysis: dipole_y_{23,24,25,26,27,28}-23-1



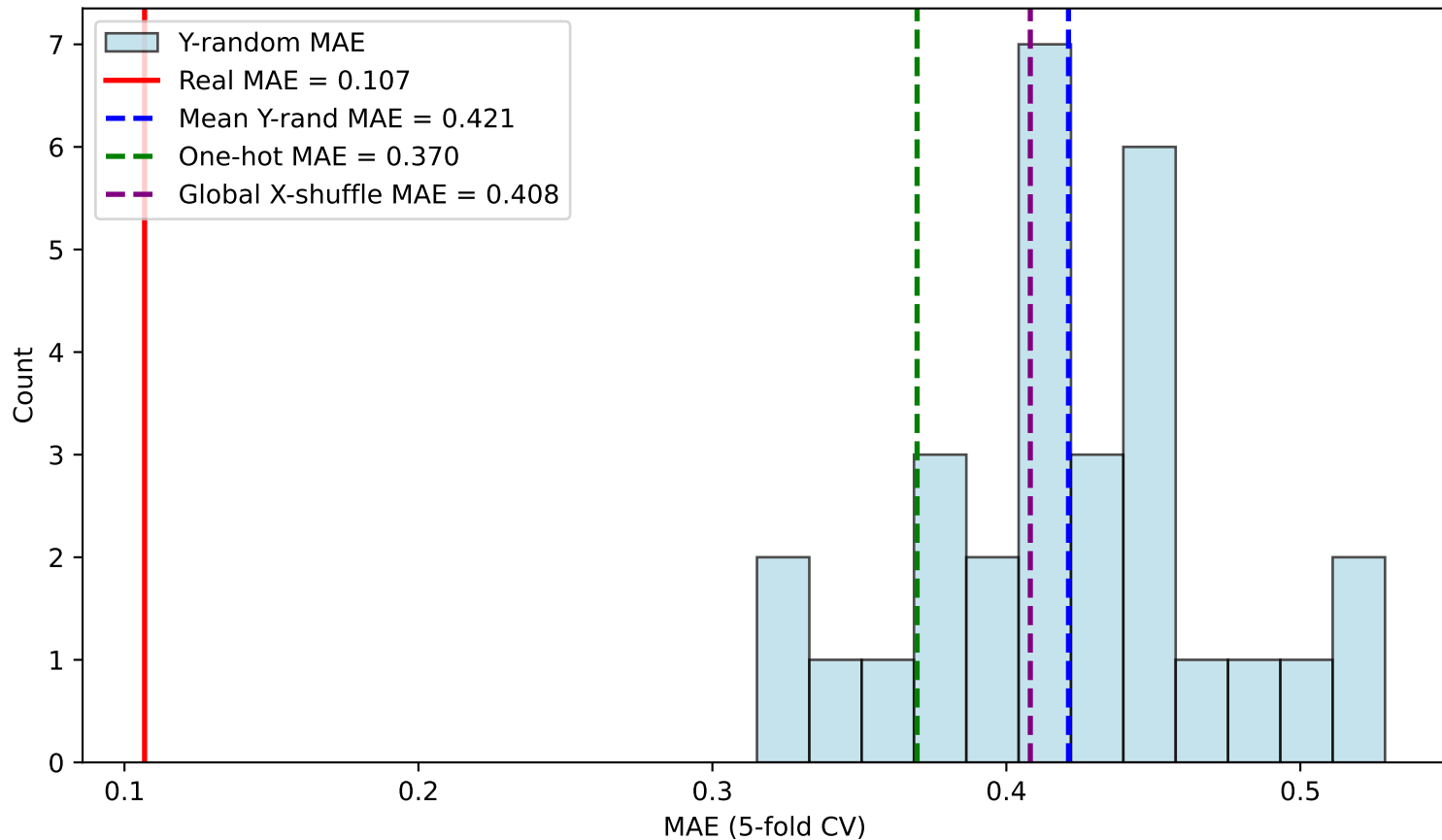
Threshold analysis: nbo_atom_3



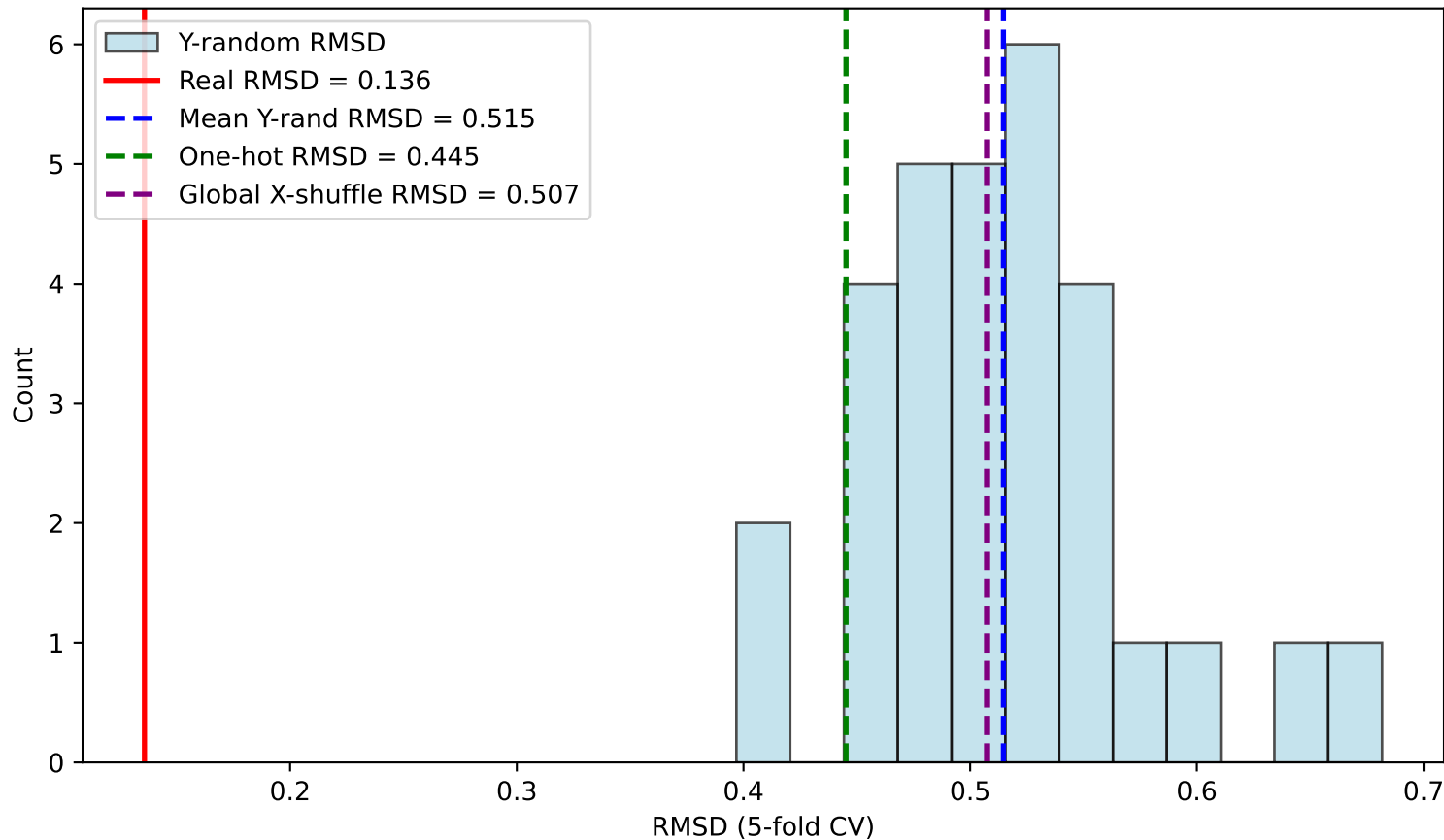
Threshold analysis: cm5_atom_1



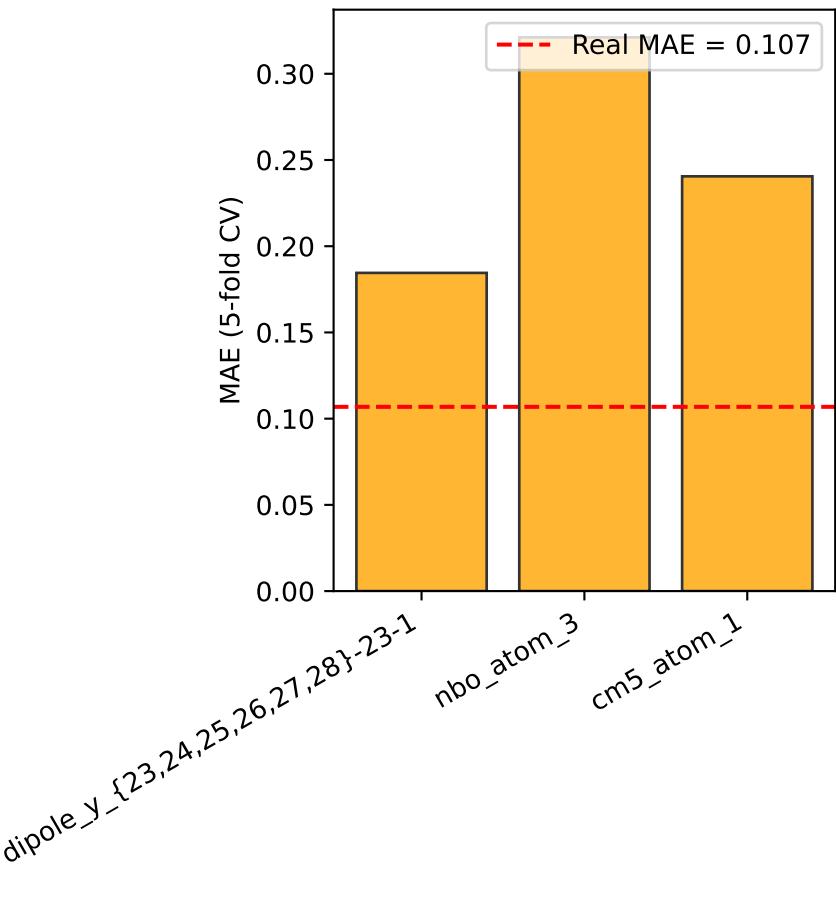
Validation (lower is better): MAE



Validation (lower is better): RMSD



Per-feature shuffle test: MAE (lower is better)



Per-feature shuffle test: RMSD (lower is b

